



JC2 Prelim Practical Exam
HIGHER 2

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9729/04

14 August 2018

2 hours 30 min

Additional Materials: As listed in your apparatus list

Write your CT group, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Lab:

For Examiner's Use	
Q1	/26
Q2	/13
Q3	/16
Total	/55

This document consists of **11** printed pages.

[Turn over

1	FA 1 is an aqueous solution prepared by mixing equal volumes of $y \text{ mol dm}^{-3}$ hydrochloric acid, HCl, and $y \text{ mol dm}^{-3}$ of sulfuric acid, H_2SO_4. FA 2 is 2.00 mol dm^{-3} sodium hydroxide, NaOH. Take care as aqueous solutions of sodium hydroxide are corrosive. When an alkali is run into an acid, an exothermic reaction takes place and the temperature of the mixture increases until the end-point is reached. If alkali is added beyond the end-point, the temperature will decrease as no further reaction takes place and the alkali is at a lower temperature than the mixture. You are to follow the neutralisation of the acids in FA 1 by measuring the temperature as known volumes of FA 2 are added.
	(a) Fill the burette to 0.00 cm^3 with FA 2. Place the Styrofoam cup in a 250 cm^3 beaker to provide support for the cup. Use the measuring cylinder to place 50.0 cm^3 of FA 1 into the cup and record the steady temperature of the acid in Table 1.1 on page 3. Read through the following instructions before starting the experiment. Record all temperatures in Table 1.1 on page 3. Run 3.00 cm^3 of FA 2 from the burette into the cup, stir the solution with the thermometer and record the maximum temperature. Immediately run a further 3.00 cm^3 of FA 2 from the burette into the cup, stir and record the maximum temperature as before. Continue the addition of FA 2, in 3.00 cm^3 portions, until 48.00 cm^3 of solution have been run from the burette.

Table 1.1

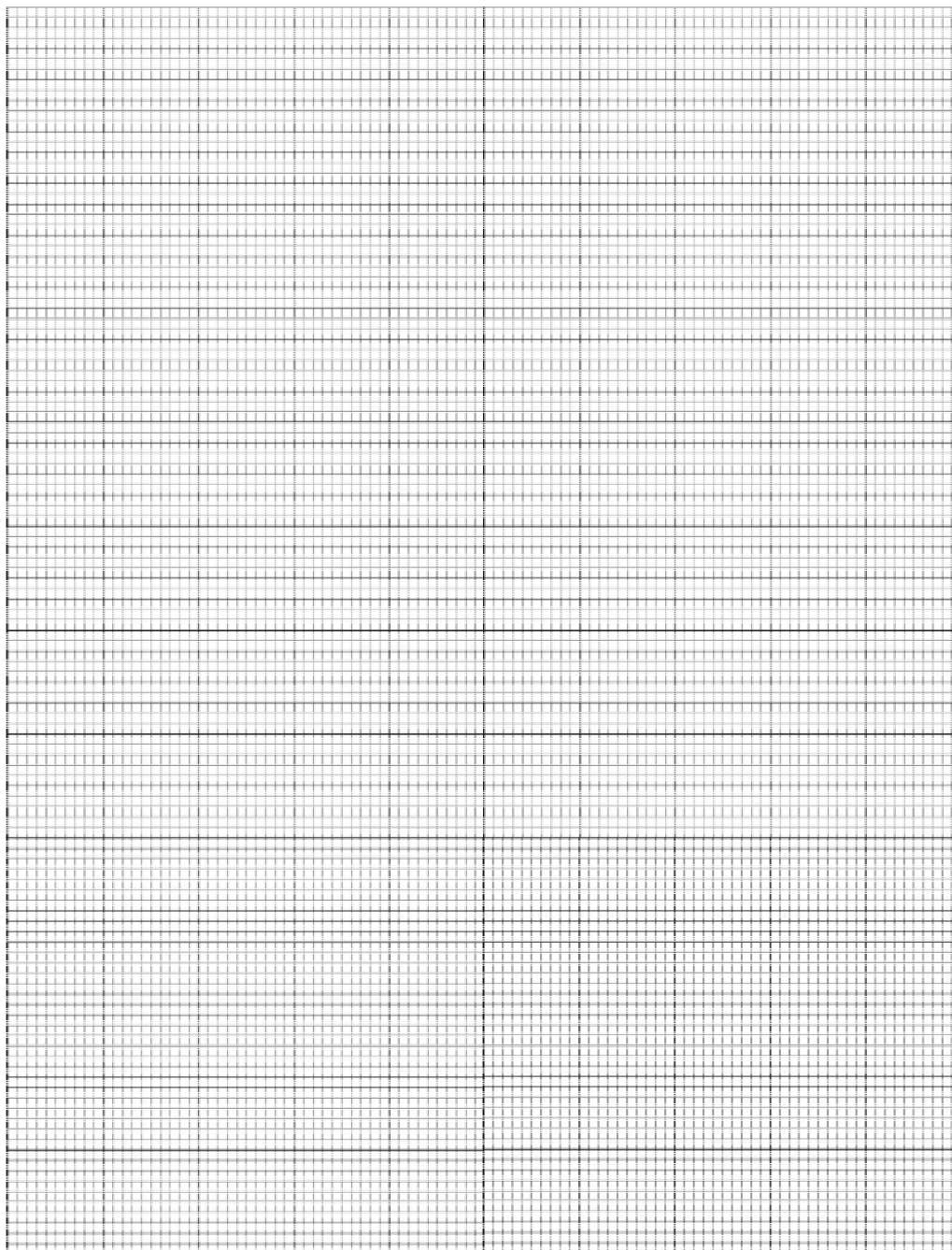
Volume of FA 2 added / cm ³	Total volume of solution in cup / cm ³	Temperature / °C		Δt ($t_x - t_0$) / °C	Total volume x Δt /
0.00	50.0	t_0	30.4	0.0	0.0
3.00	53.0	t_3	31.9		
6.00	56.0	t_6	33.3		
9.00	59.0	t_9	34.5		
12.00	62.0	t_{12}	35.6		
15.00	65.0	t_{15}	36.6		
18.00	68.0	t_{18}	37.7		
21.00	71.0	t_{21}	38.4		
24.00	74.0	t_{24}	37.8		
27.00	77.0	t_{27}	37.2		
30.00	80.0	t_{30}	36.8		
33.00	83.0	t_{33}	36.4		
36.00	86.0	t_{36}	36.0		
39.00	89.0	t_{39}	35.6		
42.00	92.0	t_{42}	35.2		
45.00	95.0	t_{45}	34.9		
48.00	98.0	t_{48}	34.6		

[3]

Complete the table by calculating Δt and (total volume x Δt) for each measurement.
Determine the units for the final column.

Units for (total volume x Δt):[1]

- (b) Plot (total volume $\times \Delta t$) against the volume of **FA 2** added.
Draw two straight lines of best-fit through the plotted points to find the end-point for the titration.



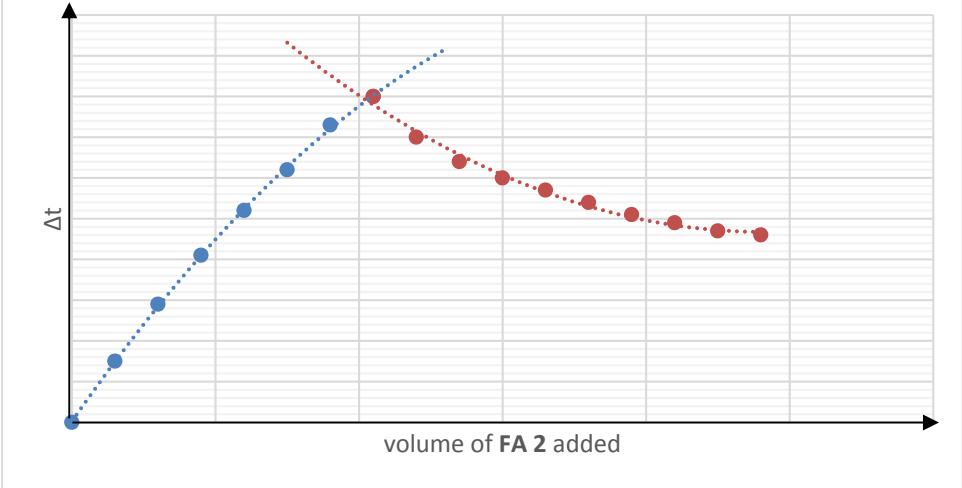
[4]

- (c) Read from the graph the end-point of the titration:20.80..... cm^3

[3]

- (d) (i) Calculate the amount of **FA 2** used to reach the end-point of the titration.

$$n_{\text{NaOH}} = 20.80 / 1000 \times 2.00 = 0.0416 \text{ mol}$$

			amount of FA 2 for end-point = [1]
	(ii)	Calculate the concentration, in mol dm ⁻³ , of the hydrogen ions in FA 1 . [1]	
		$[H^+] = 0.0416 / 0.050 = 0.832 \text{ mol dm}^{-3}$	
		concentration of H ⁺ in FA 1 = [1]	
	(iii)	Equal volumes of y mol dm ⁻³ H ₂ SO ₄ and y mol dm ⁻³ HCl were mixed to produce FA 1 . Use your answer to (d)(ii) to calculate the value of y.	
		$3y/2 = 0.832$ $y = 0.555 \text{ mol dm}^{-3}$	
		y = [1]	
	(e)	When Δt was plotted against the volume of FA 2 added, the following graph is obtained. Explain why the shape of this graph differs from that in (b).	
		Graph of Δt vs volume of FA 2 added  The heat evolved from the neutralization is spread over an increasing volume of solution, leading to a smaller Δt, hence curved lines are obtained instead.	[1]
	(f)	In the experiment you have performed, determine whether using a measuring cylinder or a burette to measure each 3.00 cm ³ portion will be more accurate.	
		Show calculations to justify your answer.	
		$\% \text{ error of measuring cylinder} = 0.5 / 3 \times 100 = \pm 16.7\%$ $\% \text{ error of burette} = 2 \times 0.05 / 3 \times 100 = \pm 3.34\%$	
		The burette would be a better apparatus as it has a smaller percentage error.	
			[5]
	(g)	Plan	

The reaction between an acid and a metal hydroxide is exothermic. The experimental procedure used in **1(a)** is known as a thermometric titration where the equivalence-point of a neutralisation reaction can be determined without the use of an indicator. This process can also be used to calculate the basicity of an acid.

A modification of the thermometric titration can be done by performing several experiments in which varied volumes of acid and base are mixed together, keeping the total volume constant. The highest temperature for each experiment is then recorded.

The data obtained can then be used to plot a graph of Δt against volume of NaOH used to determine the basicity of an acid.

Using the informaton above, plan an investigation to determine the basicity of an unknown acid H_nA of concentration 2.00 mol dm^{-3} (where $n = 1$ or 2).

You may assume that you are provided with

- 2.00 mol dm^{-3} sodium hydroxide, NaOH
- the equipment normally found in a school or college laboratory.

In your plan you should include brief details of

- the volumes of reactants that you would use,
- the apparatus that you would use
- the procedure that you would follow and the measurements that you would take,
- how you would determine the basicity of the unknown acid, H_nA via sketches of the graphs you would expect to obtain.

1. Fill up one 50 cm^3 burette with the unknown acid, and another burette with 2.00 mol dm^{-3} NaOH.

2. Using a burette, measure 10.00 cm^3 of acid into a Styrofoam cup labelled acid. Place the cup inside a second styrofoam cup which is placed in a 250 cm^3 beaker to support it.

3. Using another burette, measure out 50.00 cm^3 of NaOH into another styrofoam cup labelled FA 2.

4. Record the initial temperature of acid solution, and record the initial temperature of NaOH solution separately with a thermometer of $0.2 \text{ }^\circ\text{C}$ interval.

5. Add the NaOH(aq) from the cup containing NaOH, into the Styrofoam cup containing the acid and cover with a plastic lid.

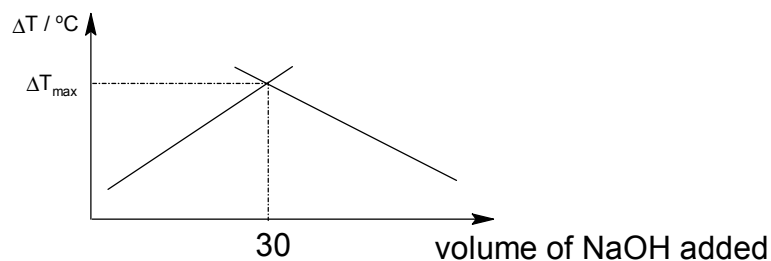
6. Stir gently the mixture with the thermometer and record the highest temperature reached

7. Wash the cups and dry them carefully with paper.

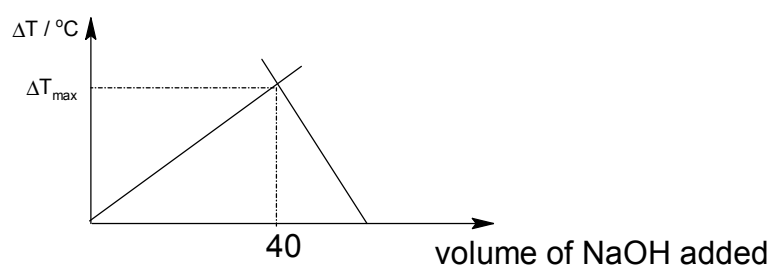
8. Repeat steps 1 to 6 for 15, 30, 40 and 50 cm³ of acid and varying volumes of NaOH(aq) to make the total volume of mixture constant at 60 cm³.

9. Sketch a graph of Δt vs volume of NaOH used.

If HA,



If H₂A,



[6]

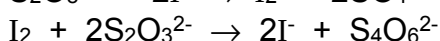
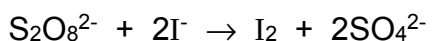
[Total: 26]

2 Determination of the rate equation for the reaction between peroxodisulfate ions and iodide ions.

FA 3 is 0.200 mol dm⁻³ aqueous ammonium peroxodisulfate, (NH₄)₂S₂O₈.

FA 4 is an aqueous solution of 1:1 mixture of 0.300 mol dm⁻³ aqueous potassium iodide, KI and 0.00800 mol dm⁻³ aqueous sodium thiosulfate, Na₂S₂O₃.

If potassium iodide and ammonium peroxodisulfate are mixed in the presence of starch and sodium thiosulfate, the iodine liberated reacts with the thiosulfate ions until they are all used up.



When no thiosulfate ions are left, any iodine formed reacts with starch and the dark blue colour of the starch-iodine complex appears. The time taken for the dark blue colour to appear with a constant quantity of sodium thiosulfate present depends on the rate of formation of iodine

You are required to carry out a total of **four** experiments at different concentration of peroxodisulfate ions and note the time taken, **t**, for the solutions to turn dark blue.

The end-point of the reaction is the first appearance of a dark blue colour.

You need to swirl the reaction mixture well as insufficient swirling will lead to a dark blue colour appearing before the true end-point is reached. Keep to the same number of swirls in each experiment.

Prepare a table in the space provided on **page 10** in which to record, for each experiment:

- volume of **FA 3**
- volume of deionised water
- values of **t**
- calculated values of **1/t**

*Note: As the volume of **FA 4** is constant for each experiment, you do not need to record its volume in your table.*

You are advised to read the procedure and to prepare a table for your results before starting the practical work.

(a)	Procedure		
	Experiment 1		
	1.	Fill the burette labelled FA 3 with FA 3 .	
	2.	Run 20.00 cm ³ of FA 3 into a 100 cm ³ beaker.	
	3.	Using a 25 cm ³ measuring cylinder, measure out 20 cm ³ of FA 4 and add 10 drops of starch solution.	

		4.	Add the contents of the measuring cylinder to the beaker and start the stopwatch when half of the contents has been added.
		5.	Swirl the mixture and place the beaker on a white tile.
		6.	Stop the stopwatch as soon as the dark blue colour appears. Record the time, t , to the nearest second.
		7.	Wash both the measuring cylinder and the beaker with water and wipe dry.
		Experiment 2	
		1.	Run 10.00 cm ³ of FA 3 into a 100 cm ³ beaker.
		2.	Using a 25 cm ³ measuring cylinder, measure out 10 cm ³ of deionised water into the beaker containing FA 3 .
		3.	Using another 25 cm ³ measuring cylinder, measure out 20 cm ³ of FA 4 and add 10 drops of starch solution.
		4.	Add the contents of the measuring cylinder to the beaker and start the stopwatch when half of the contents has been added.
		5.	Swirl the mixture and place the beaker on a white tile.
		6.	Stop the stopwatch as soon as the dark blue colour appears. Record the time, t , to the nearest second.
		7.	Wash both the measuring cylinder and the beaker with water and wipe dry.
		Experiment 3 – 4	
		Perform two additional experiments using different volume of the peroxodisulfate solution and the deionised water. Note that the combined volume of FA 3 and deionised water must always be 20 cm ³ . Do not use a volume of FA 3 that is less than 6 cm ³ .	
	(b)	In the series of experiments where the total volume of the mixture is kept constant and the same end-point (appearance of dark blue colour) is timed at different concentrations of peroxodisulfate solution, • $[\text{S}_2\text{O}_8^{2-}] \propto V_{\text{FA3 used}}$ • $\text{rate} \propto \frac{1}{t}$ Record the results of your four experiments below. Table of results	

Experiment	Volume of FA3/ cm ³	Volume of deionised water/ cm ³	t / s	1/t s ⁻¹
1	20.00	0.0	23	
2	10.00	10.0	49	
3	17.00	3.0	27	
4	13.00	7.0	31	
5	7.00	13.0	71	

[5]

(c)	Using the results in your table, deduce the order of the reaction with respect to the peroxodisulfate ions. Explain clearly how you arrived at your answer.			
	Order is 1 . **Comparing expt 1 and 2,** **Halve [S₂O₈²⁻], time is doubled**			
	[1]			
(d)	(i)	Based on experiments 1 – 4, indicate, with reasoning, which experiment is most likely to have the largest error in the time measured.		
		The time measured for experiment 1 is the shortest and hence have the greatest percentage error.		
		[1]		
	(ii)	Instead of washing the beaker as required before performing the next experiment, a student simply just poured away the reaction mixture. There were some left over reaction mixture in the beaker when he performed the subsequent experiment. State and explain what will happen to the time, t.		
		The excess I₂ left in the flask will react with the S₂O₃²⁻ in the new experiment. As there is lesser S₂O₃²⁻ for the new experiment, less I₂ is reacted. Given that the rate of reaction is the same, the time taken to produce the I₂ will be shorter and hence observe the blue colour faster.		
		[2]		
(e)	(i)	Explain why a fixed amount of sodium thiosulfate is required.		

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3 FA 5 contains Ni^{2+} and one other cation and one anion.

In all tests, the reagents should be added gradually until no further change is observed after shaking with each addition.

Record your observations and the deductions you make from them in the spaces provided

Your answers should include

- Details of colour changes and precipitates formed
- The names of gases evolved and details of the test used to identify each one.

You should indicate clearly at what stage in a test, an observation occurs

No additional or confirmatory tests for ions present should be attempted

Test	Observations
(i) Test the FA 5 solution with Universal Indicator paper.	pH is 3
(ii) To 2 cm depth of FA 5 in a test tube, add aqueous sodium carbonate until no further change is observed.	green ppt Effervescence seen , gas forms white ppt when passes through limewater
(iii) To 3 cm depth of FA 5 in a boiling tube, add aqueous sodium hydroxide until no further change is observed. Filter the mixture and collect the filtrate, label it FA 6 . State the colour of the residue and filtrate. Divide FA 6 into two portions in two separate test tubes for use in (iv) and (v).	Green ppt <u>insoluble</u> in excess aqueous sodium hydroxide (light) Green residue and Colourless filtrate
(iv) To 2 cm depth of FA 6 in a test-tube, add dilute nitric acid, drop by drop, until no further change is observed.	No effervescence / no gas evolved (Thin cloudy) White ppt appears which is soluble in excess nitric acid
(v) To 2 cm depth of FA 6 add aqueous silver nitrate.	White ppt forms
Add an excess of dilute aqueous ammonia.	<u>soluble</u> in excess aqueous ammonia

[5]

Conclusion		
(a)	Consider your observations in (ii). Suggest an explanation for the evolution of gas observed. Write an equation, including state symbols in your answer.	
	FA 5 contains an acidic salt which undergoes hydrolysis to produce H^+ . The H^+ would react with carbonate to produce CO_2 $2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	
	[2]	
(b)	Consider your observations in (v). Suggest an explanation for the observations in terms of the chemistry involved.	
	Cl^- reacts with AgNO_3 to form white ppt of AgCl or show equation With $\text{NH}_3(\text{aq})$, complex $[\text{Ag}(\text{NH}_3)_2]^+$ is formed or show equation . This reduces $[\text{Ag}^+]$ and shifts the equilibrium: $\text{AgCl}(\text{s}) \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$ to the right OR ionic product becomes lower than K_{sp} , so AgCl solid dissolves.	
	[2]	
(c)	(i)	FA 5 contains the cations: Ni^{2+} and Al^{3+}
		[1]
	(ii)	FA 5 contains the anion: Cl^-
		[1]
(d)	Plan FA 7 is a solution that contains Fe^{2+} and Al^{3+} ions Plan a sequence of steps by which the two ions could be separated so that each ion is present in a separate precipitate. For each of the steps, you are to specify the location of the ions and their formulae. You are also required to specify the amount (dropwise or excess) of reagents that you add. You may assume that the usual bench reagents and apparatus are available for use.	
	1. Add excess $\text{NaOH}(\text{aq})$ to the mixture. Carry out filtration to separate the insoluble ppt / $\text{Fe}(\text{OH})_2$ from the solution Residue: $\text{Fe}(\text{OH})_2(\text{s})$ Filtrate: $\text{Al}(\text{OH})_4^-(\text{aq})$ 2. To the filtrate[1], Add HCl dropwise 3. Carry out filtration to separate the insoluble ppt/ $\text{Al}(\text{OH})_3(\text{s})$ from the solution Residue: $\text{Al}(\text{OH})_3$	

				[5]
				[Total: 16]